

Computer vision uses algorithms to extract, analyze, and interpret meaningful information from digital images and videos, mimicking human visual understanding to identify objects, patterns, and scenes.

(a) Main objectives of computer vision: To emulate human vision capabilities, enable object and scene recognition, extract useful information from images, and support automated decision-making.

(b) Two real-world applications: Facial recognition (used for security and unlocking devices) and medical image diagnosis (detecting tumors in scans).

2) (a) Feature extraction: The process of identifying and isolating distinctive attributes - such as edges, corners, and textures - from an image that help describe or classify it.

(b) How object recognition improves systems: It allows systems to automatically identify and label objects, enabling tasks like classification, tracking, and decision-making without manual input.

3) (a) Low-level Image preprocessing: Operations performed directly on raw pixel data, such as noise reduction and contrast enhancement, where both the input and output are images.

(b) Mid-level vs High-level Image preprocessing: Mid-level preprocessing takes an image as input and produces attributes or features as output.

High-level preprocessing interprets these features to make sense of the scene.

4) (a) Pixel: The smallest unit of a digital image, representing a single point with a specific intensity or color value.

(b) Effect of Image resolution: Resolution determines the level of detail in an image; higher resolution means sharper images with finer detail, while lower resolution causes blurriness or pixelation.

- 5) (a) Role of an Image sensor: It converts light reflected from a scene into electrical signals, which are then digitized into pixel values.
- (b) Two image acquisition devices: Digital cameras and scanners (CCD/CMOS sensors and MRI machines are also acceptable examples).

6) (a) Sampling: The Process of converting a continuous image into a discrete grid of pixels by measuring intensity at fixed spatial intervals.

(b) Purpose of quantization: To assign discrete numerical intensity or color values to each sampled point, converting continuous brightness levels into a finite set of digital values.

7) (a) Perspective projection: The geometric process of mapping 3D world points onto a 2D image plane, where objects farther away appear smaller, similar to how cameras and eyes perceive the world.

(b) Role of illumination: The intensity and direction of the light source affect how objects reflect light, influencing the brightness, shadows, and contrast captured in the image.

8) (a) Image preprocessing: The manipulation of an image to enhance it or extract basic information, where the output is another image.

(b) Two differences: (i) Image preprocessing produces an image as output, while computer vision produces interpretation or understanding, such as labels or decisions. (ii) Image preprocessing focuses on low-level pixel operations, while computer vision focuses on high-level understanding and reasoning about content.

(a) Application in healthcare: Computer vision analyzes medical scans, such as x-rays and MRIs, to detect abnormalities like tumors, helping doctors make faster and more accurate diagnoses.

(b) Application in autonomous vehicles: It enables self-driving cars to detect lanes, pedestrians, traffic signs, and obstacles in real time, supporting safe navigation.

(a) Why pixels are the basic building blocks: Every visual feature in an image - such as edges, colors, and shapes - is ultimately derived from analyzing pixel values and their relationships to neighboring pixels.

(b) How sampling and resolution influence images: Sampling determines how much spatial detail is captured, and resolution affects clarity. Higher sampling and resolution provide better

data for accurate feature extraction, while
low resolution can hinder object detection and
recognition accuracy.

